Simple linear regression. Simple linear regression models the relationship between one predictor x and a numeric response y with the line y = b0 + b1*x + error. The coefficients are estimated by ordinary least squares, which chooses the line that minimises the sum of squared residuals. A residual is the difference between an observed value and the value predicted by the line. The coefficient of determination, R-squared, is the proportion of variance in the response explained by the model and always lies between 0 and 1 for a model with an intercept. Simple regression has exactly one predictor, so it cannot separate the effects of several correlated variables.

Multiple regression. Multiple regression extends the model to several predictors: y = b0 + b1*x1 + b2*x2 + ... + bk*xk + error. Each coefficient is interpreted as the expected change in the response for a one unit increase in that predictor while all other predictors are held constant. Adding predictors never decreases R-squared, so adjusted R-squared is used because it penalises predictors that add no explanatory power. Multicollinearity occurs when predictors are strongly correlated with each other; it inflates the standard errors of the coefficients. The variance inflation factor, VIF, quantifies it, and a VIF above 10 is a common warning sign.

Overfitting and regularisation. A model overfits when it fits noise in the training data and performs worse on new data. The bias-variance trade-off describes how simple models have high bias and low variance while flexible models have low bias and high variance. Ridge regression adds an L2 penalty equal to the sum of squared coefficients, which shrinks coefficients towards zero but rarely makes them exactly zero. Lasso regression adds an L1 penalty equal to the sum of absolute coefficients, which can set some coefficients exactly to zero and therefore performs feature selection. K-fold cross-validation estimates generalisation error by training on k-1 folds and testing on the remaining fold, repeating k times.